Pragmastat: Pragmatic Statistical Toolkit

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Abstract

This manual presents a unified statistical toolkit for reliable analysis of real-world data. The toolkit nearly matches the efficiency of traditional statistical estimators under normality, maintains reasonable robustness against outliers, enables simple software implementations without advanced statistical libraries, and provides clear explanations accessible to practitioners without deep statistical training. The toolkit consists of renamed, recombined, and refined versions of existing methods.

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1 Toolkit Overview

1.1 Introduction

This toolkit analyzes random variables X and Y. Each random variable produces random real numbers from its underlying distribution. Consider a set of n random values $\mathbf{X}$ and a set of m random values $\mathbf{Y}$:

$$\mathbf{X} = (X_1, X_2, \ldots, X_n)$$

$$\mathbf{Y} = (Y_1, Y_2, \ldots, Y_m)$$

These variables produce *measurements* (the actual observed values). For random values $\mathbf{X}$, the measurements are $x_1, x_2, \ldots, x_n$. This collection of measurements forms a *sample* $\mathbf{x}$:

$$\mathbf{x} = (x_1, x_2, \ldots, x_n)$$

$$\mathbf{y} = (y_1, y_2, \ldots, y_m)$$

The objective is to learn about the original random variables X and Y from these samples.

The *median* represents a fundamental property of any distribution. This value divides the distribution into two equal parts, creating exactly 50% probability of observing a measurement below the median and 50% above it. The *true* median value $\text{Median}(X)$ remains unknown, but the measurements provide an estimate.

To calculate the median, arrange the sample elements into an ordered sequence $x_{(1)}, \ldots, x_{(n)}$. This *sorted sample*[1] places $x_{(1)}$ as the smallest value and $x_{(n)}$ as the largest. The median estimate uses the middle value of this sorted sample. For even sample sizes, the median equals the average of the two middle values:

$$\boxed{\text{Median}(\mathbf{x}) = \begin{cases} x_{((n+1)/2)} & \text{if } n \text{ is odd} \\ \frac{x_{(n/2)} + x_{(n/2+1)}}{2} & \text{if } n \text{ is even} \end{cases}}$$

This estimation $\text{Median}(\mathbf{x})$ approximates the true distribution value $\text{Median}(X)$.

The median represents one example of a *quantile*. For probability $p \in [0, 1]$, the pth quantile $\text{Quantile}(X, p)$ divides the distribution so that the probability of obtaining a measurement below the quantile equals exactly p:

$$\Pr[X \leq \text{Quantile}(X, p)] = p$$

Linear interpolation between two sample elements produces quantile estimates[2]:

$$\boxed{\begin{gathered} \text{Quantile}(\mathbf{x}, p) = x_{(\lfloor h \rfloor)} + (h - \lfloor h \rfloor) \cdot (x_{(\lceil h \rceil)} - x_{(\lfloor h \rfloor)}) \\ h = (n-1)p + 1 \end{gathered}}$$

The median equals $\text{Quantile}(\mathbf{x}, 0.5)$ since it divides the distribution into equal halves. Median estimation produces higher accuracy than other quantile values. Estimation accuracy decreases as p approaches 0 or 1. Larger sample sizes improve quantile estimation accuracy for any value of p.

[1] Also known as *order statistics*

[2] Definition uses the Hyndman-Fan Type 7 quantile estimator, see [HF96].

These quantile estimators form the building blocks for more complex distributional properties.

The following sections examine one-sample and two-sample estimators that provide practical insights about data. Estimator evaluation focuses on these key properties:

- *Gaussian Efficiency*: performance quality under normal distribution conditions. High efficiency produces accurate estimates with fewer measurements under normality.
- *Robustness*: stability when samples contain extreme values (outliers). High robustness maintains accuracy even with contaminated data.

1.2 One-Sample Summary

$$\boxed{\mathrm{Center}(\mathbf{x}) = \underset{1 \le i \le j \le n}{\mathrm{Median}} \left(\frac{x_i + x_j}{2} \right)}$$

The *center*[3] provides the primary insight into any sample (consistent with the median for symmetric distributions). This estimator excels in problems requiring central tendency measures.

Compared to the median, the center produces higher efficiency under normality and requires 1.5 times fewer observations for equivalent precision. Compared to the arithmetic mean, the center offers greater robustness (tolerates nearly one-third outliers) while maintaining comparable efficiency under normality.

$$\boxed{\mathrm{Spread}(\mathbf{x}) = \underset{1 \le i < j \le n}{\mathrm{Median}} |x_i - x_j|}$$

The *spread*[4] provides the second key insight into sample characteristics by measuring dispersion (variability or scatter). This estimator $\mathrm{Spread}(\mathbf{x})$ calculates the median absolute difference between all pairs of measurements. Under normal distribution conditions, multiplication by a constant converts spread to other dispersion measures.

Compared to standard deviation, spread provides greater robustness (tolerates nearly one-third outliers) with comparable efficiency under normality. Compared to median absolute deviation, spread produces higher efficiency under normality and requires 2.35 times fewer observations for equivalent precision.

$$\boxed{\mathrm{RelSpread}(\mathbf{x}) = \frac{\mathrm{Spread}(\mathbf{x})}{|\mathrm{Center}(\mathbf{x})|}}$$

The relative spread $\mathrm{RelSpread}(\mathbf{x})$ measures dispersion normalized by the center value. Percentage expression proves convenient: 0.2 represents 20% relative to $\mathrm{Center}(\mathbf{x})$. Scale invariance makes RelSpread useful for portable experimental designs.

[3] Also known as the *Hodges–Lehmann* location estimator or pseudomedian, see [HL63], [Sen63]
[4] Also known as the *Shamos* scale estimator, see [Sha76]

1.3 Two-Sample Summary

The Shift expresses the median difference between two random variables:

$$\text{Shift}(X, Y) = \text{Median}(X - Y)$$

When taking measurements from X and Y, the difference falls below $\text{Shift}(X, Y)$ with 50% probability and above it with 50% probability. Identical distributions produce zero shift: $\text{Shift}(X, X) = 0$.

The estimator for the shift[5] is defined as follows:

$$\boxed{\text{Shift}(\mathbf{x}, \mathbf{y}) = \underset{1 \le i \le n,\ 1 \le j \le m}{\text{Median}} (x_i - y_j)}$$

This estimator calculates the median difference between elements of $\mathbf{x}$ and $\mathbf{y}$. Sign interpretation proves critical: positive shifts indicate larger $\mathbf{x}$ values, negative shifts indicate larger $\mathbf{y}$ values. Small values of $\text{Spread}(X)$ and $\text{Spread}(Y)$ make $\text{Shift}(\mathbf{x}, \mathbf{y})$ closely approximate the difference between any x_i and y_j.

Absolute units for expressing differences between X and Y create problems and reduce experimental portability. Relative measures often provide better solutions:

$$\text{Ratio}(X, Y) = \text{Median}\left(\frac{X}{Y}\right)$$

The $\text{Ratio}(X, Y)$ captures the median ratio between random measurements from X and Y:

$$\boxed{\text{Ratio}(\mathbf{x}, \mathbf{y}) = \underset{1 \le i \le n,\ 1 \le j \le m}{\text{Median}} \left(\frac{x_i}{y_j}\right)}$$

This estimator $\text{Ratio}(\mathbf{x}, \mathbf{y})$[6] calculates the median ratio of $\mathbf{x}$ elements to $\mathbf{y}$ elements. For example, Ratio = 1.2 indicates that the median ratio is 1.2, meaning 50% of (x_i, y_j) pairs have ratios below 1.2 and 50% above 1.2. Percentage expression follows $(\text{Ratio} - 1) \times 100\%$. Scale invariance makes ratio estimation useful for portable experimental designs.

Neither shift nor ratio works universally well. Portable experimental design benefits from expressing absolute differences in scale-invariant units. This requirement introduces *disparity*:

$$\text{Disparity}(X, Y) = \frac{\text{Median}(X - Y)}{\text{AvgSpread}(X, Y)}$$

where

$$\text{AvgSpread}(X, Y) = \frac{\text{Spread}(X) + \text{Spread}(Y)}{2}$$

[5] Also known as the *Hodges–Lehmann shift estimator*
[6] Inspired by the *Hodges–Lehmann estimator*

The Disparity(X, Y) normalizes the shift by spread, expressing differences in scale-invariant spread units.

$$\boxed{\text{Disparity}(\mathbf{x}, \mathbf{y}) = \frac{\text{Shift}(\mathbf{x}, \mathbf{y})}{\text{AvgSpread}(\mathbf{x}, \mathbf{y})}}$$

where

$$\text{AvgSpread}(\mathbf{x}, \mathbf{y}) = \frac{n \cdot \text{Spread}(\mathbf{x}) + m \cdot \text{Spread}(\mathbf{y})}{n + m}$$

2 Estimators

This section explains each estimator's key properties, selection rationale, and advantages over traditional methods.

2.1 Center

$$\text{Center}(\mathbf{x}) = \underset{1 \le i \le j \le n}{\text{Median}} \left(\frac{x_i + x_j}{2} \right)$$

Practical Recommendations

The Center provides initial insight into the magnitude of sample values. When RelSpread is small, the entire sample can be approximated by the Center value.

Key Facts

- Measures central tendency (the average value)
- Domain: any real numbers
- Equals the *Hodges-Lehmann estimator* ([HL63], [Sen63]), renamed to "Center" for clarity
- Called *pseudomedian* in some texts because it is consistent with the median for symmetric distributions
- Asymptotic Gaussian efficiency: $\approx 96\%$; finite-sample Gaussian efficiency $> 91\%$
- Asymptotic breakdown point: $\approx 29\%$

Comparison

- Compared to the *mean*: more robust (tolerates almost one-third of outliers) and has comparable efficiency under normality
- Compared to the *median*: more efficient under normality and requires 1.5 times fewer observations for the same precision

Properties

$$\text{Center}(\mathbf{x} + k) = \text{Center}(\mathbf{x}) + k$$

$$\text{Center}(k \cdot \mathbf{x}) = k \cdot \text{Center}(\mathbf{x})$$

2.2 Spread

$$\text{Spread}(\mathbf{x}) = \underset{i<j}{\text{Median}}|x_i - x_j|$$

Practical Recommendations

The Spread provides initial insight into the dispersion of the sample values. Interpretation: half of $|x_i - x_j|$ values are smaller than $\text{Spread}(\mathbf{x})$, the other half are larger.

Key Facts

- Measures dispersion (also known as variability or scatter)
- Domain: any real numbers (for $n = 1$, use $\text{Spread}(\mathbf{x}) = 0$)
- Equals the *Shamos estimator* ([Sha76]), renamed to "Spread" for clarity
- Asymptotic Gaussian efficiency: $\approx 86\%$
- Asymptotic breakdown point: $\approx 29\%$ (matches Center in robustness)
- Asymptotic expected value for the standard normal distribution: ≈ 0.954
- Not consistent with the standard deviation under normality

Comparison

- Compared to the *standard deviation*: more robust (tolerates almost one-third of outliers) and has comparable efficiency under normality; more intuitive without requiring knowledge of normal distributions
- Compared to the *median absolute deviation*: more efficient under normality and requires approximately 2.35 times fewer observations for the same precision

Empirical Rule

For the standard normal distribution, the asymptotic 68–95–99.7 rule becomes the 66–94–99.6 rule:

- $[\text{Center}(\mathbf{x}) \pm 1 \cdot \text{Spread}(\mathbf{x})]$ covers $\approx 65.98518\%$ of the distribution
- $[\text{Center}(\mathbf{x}) \pm 2 \cdot \text{Spread}(\mathbf{x})]$ covers $\approx 94.35758\%$ of the distribution
- $[\text{Center}(\mathbf{x}) \pm 3 \cdot \text{Spread}(\mathbf{x})]$ covers $\approx 99.57851\%$ of the distribution
- $[\text{Center}(\mathbf{x}) \pm 4 \cdot \text{Spread}(\mathbf{x})]$ covers $\approx 99.98641\%$ of the distribution
- $[\text{Center}(\mathbf{x}) \pm 5 \cdot \text{Spread}(\mathbf{x})]$ covers $\approx 99.99982\%$ of the distribution

Properties

$$\text{Spread}(\mathbf{x} + k) = \text{Spread}(\mathbf{x})$$

$$\text{Spread}(k \cdot \mathbf{x}) = |k| \cdot \text{Spread}(\mathbf{x})$$

$$\text{Spread}(\mathbf{x}) \geq 0$$

2.3 RelSpread

$$\text{RelSpread}(\mathbf{x}) = \frac{\text{Spread}(\mathbf{x})}{|\text{Center}(\mathbf{x})|}$$

Practical Recommendations

The RelSpread provides scale-invariant insight into the distribution dispersion normalized by the center value.

Interpretation examples:

- $\text{RelSpread}(\mathbf{x}) = 1\%$: data clusters tightly around $\text{Center}(\mathbf{x})$ with minimal variation
- $\text{RelSpread}(\mathbf{x}) = 10\%$: moderate variation, typical values range from 90% to 110% of center
- $\text{RelSpread}(\mathbf{x}) = 100\%$: high variation, values span from near zero to twice the center

Key Facts

- Measures the relative dispersion of a sample to $\text{Center}(\mathbf{x})$
- Domain:
  - Mathematical Domain: $\text{Center}(\mathbf{x}) \neq 0$
  - Logical Domain: all sample elements have the same sign, sample doesn't contain zeros
  - Pragmatic Domain: non-negative values allowing up to 29% zeros
- Robust alternative to the *coefficient of variation*
- Scale-invariant, which makes experimental design more portable

Properties

$$\text{RelSpread}(k \cdot \mathbf{x}) = \text{RelSpread}(\mathbf{x})$$

$$\text{RelSpread}(\mathbf{x}) \geq 0$$

2.4 Shift

$$\text{Shift}(\mathbf{x}, \mathbf{y}) = \underset{1 \leq i \leq n,\ 1 \leq j \leq m}{\text{Median}} (x_i - y_j)$$

Practical Recommendations

The Shift provides initial insight into the difference between elements of two samples. Interpretation: half of $x_i - y_j$ values are smaller than $\text{Shift}(\mathbf{x}, \mathbf{y})$, the other half are larger. For samples with small RelSpread, $\text{Shift}(\mathbf{x}, \mathbf{y})$ approximates pairwise differences $x_i - y_j$.

Key Facts

- Measures the median difference between elements of two samples
- Domain: any real numbers
- Equals the *Hodges-Lehmann estimator* for two samples ([HL63])

Properties

$$\text{Shift}(\mathbf{x} + k_x, \mathbf{y} + k_y) = \text{Shift}(\mathbf{x}, \mathbf{y}) + k_x - k_y$$

$$\text{Shift}(k \cdot \mathbf{x}, k \cdot \mathbf{y}) = k \cdot \text{Shift}(\mathbf{x}, \mathbf{y})$$

$$\text{Shift}(\mathbf{x}, \mathbf{y}) = -\,\text{Shift}(\mathbf{y}, \mathbf{x})$$

2.5 Ratio

$$\text{Ratio}(\mathbf{x}, \mathbf{y}) = \underset{1 \le i \le n,\ 1 \le j \le m}{\text{Median}} \left(\frac{x_i}{y_j} \right)$$

Practical Recommendations

The Ratio provides initial insight into the ratio between elements of two samples expressed as a multiplicative factor. It answers "how many times larger is $\mathbf{x}$ compared to $\mathbf{y}$?" For example, Ratio = 2.0 means that for 50% of pairs (x_i, y_j), x_i is at least twice as large as y_j.

The Ratio functions as a division operator: $\text{Ratio}(\mathbf{x}, \mathbf{y})$ computes the typical ratio $\mathbf{x}/\mathbf{y}$. Results convert to percentage differences as $(\text{Ratio} - 1) \times 100\%$.

Key Facts

- Measures the median ratio between elements of two samples
- Domain: $y_j > 0$
- Second sample y is always the baseline
- In general, $\text{Ratio}(\mathbf{x}, \mathbf{y}) \neq 1/\,\text{Ratio}(\mathbf{y}, \mathbf{x})$ (for example, $x = (1, 100)$, $y = (1, 10)$)

Properties

$$\text{Ratio}(k_x \cdot \mathbf{x}, k_y \cdot \mathbf{y}) = \frac{k_x}{k_y} \cdot \text{Ratio}(\mathbf{x}, \mathbf{y})$$

2.6 AvgSpread

$$\text{AvgSpread}(\mathbf{x}, \mathbf{y}) = \frac{n\,\text{Spread}(\mathbf{x}) + m\,\text{Spread}(\mathbf{y})}{n + m}$$

Practical Recommendations

The AvgSpread primarily serves as a scaling factor for Disparity. It represents the combined dispersion of both samples, weighted by sample size. Performs best for distributions with similar dispersion values.

Key Facts

- Measures average dispersion across two samples
- Domain: any real numbers
- Provides robust alternative to the pooled standard deviation

Properties

$$\text{AvgSpread}(\mathbf{x}, \mathbf{x}) = \text{Spread}(\mathbf{x})$$

$$\text{AvgSpread}(k_1 \cdot \mathbf{x}, k_2 \cdot \mathbf{x}) = \frac{|k_1| + |k_2|}{2} \cdot \text{Spread}(\mathbf{x})$$

$$\text{AvgSpread}(\mathbf{x}, \mathbf{y}) = \text{AvgSpread}(\mathbf{y}, \mathbf{x})$$

$$\text{AvgSpread}(k \cdot \mathbf{x}, k \cdot \mathbf{y}) = |k| \cdot \text{AvgSpread}(\mathbf{x}, \mathbf{y})$$

2.7 Disparity

$$\text{Disparity}(\mathbf{x}, \mathbf{y}) = \frac{\text{Shift}(\mathbf{x}, \mathbf{y})}{\text{AvgSpread}(\mathbf{x}, \mathbf{y})}$$

Practical Recommendations

The Disparity provides scale-invariant insight into the difference between elements of two samples, expressed in standardized spread units.

Key Facts

- Measures a normalized difference between $\mathbf{x}$ and $\mathbf{y}$ expressed in standardized spread units
- Domain: $\text{AvgSpread}(\mathbf{x}, \mathbf{y}) > 0$ (at least 50% of $|x_i - x_j|$ and 50% of $|y_i - y_j|$ are non-zero)
- Expresses the *effect size*, renamed to "Disparity" for clarity
- Scale-invariant, which makes experimental design more portable

Comparison

- Compared to *Cohen's d*: more robust while maintaining efficiency under normality

Properties

$$\text{Disparity}(\mathbf{x} + k, \mathbf{y} + k) = \text{Disparity}(\mathbf{x}, \mathbf{y})$$

$$\text{Disparity}(k \cdot \mathbf{x}, k \cdot \mathbf{y}) = \text{sign}(k) \cdot \text{Disparity}(\mathbf{x}, \mathbf{y})$$

$$\text{Disparity}(\mathbf{x}, \mathbf{y}) = -\,\text{Disparity}(\mathbf{y}, \mathbf{x})$$

3 Studies

This section analyzes the estimators' properties using mathematical proofs and Monte Carlo simulations. Most proofs are adopted from various textbooks and research papers, but only essential references are provided.

Unlike the main part of the manual, the studies require knowledge of classic statistical methods. Well-known facts and commonly accepted notation are used without special introduction. The studies provide detailed analysis of estimator properties for practitioners interested in rigorous proofs and numerical simulation results.

3.1 Asymptotic Gaussian Expected Value of the Spread

This study establishes that Spread has expected value $\sqrt{2}\,\Phi^{-1}(0.75) \approx 0.954$ under standard normal data as sample size increases.

The key insight: pairwise absolute differences $|X_i - X_j|$ from normal data converge to a known distribution. Since Spread takes the median of these differences, its asymptotic expectation equals the population median of $|X_1 - X_2|$ where $X_1, X_2 \overset{\text{iid}}{\sim} \mathcal{N}(0,1)$.

Consider $X_1, \ldots, X_n \overset{\text{iid}}{\sim} \mathcal{N}(0,1)$. For any fixed $i \neq j$, the difference $X_i - X_j$ has mean 0 and variance

$$\mathbb{V}[X_i - X_j] = \mathbb{V}[X_i] + \mathbb{V}[X_j] = 1 + 1 = 2$$

because X_i and X_j are independent. Therefore

$$X_i - X_j \sim \mathcal{N}(0,2)$$

Define $W = (X_i - X_j)/\sqrt{2} \sim \mathcal{N}(0,1)$ and $D = |X_i - X_j| = \sqrt{2}\,|W|$. The cumulative distribution function of D for $t \geq 0$ is

$$\Pr[D \leq t] = \Pr\big[|W| \leq t/\sqrt{2}\big] = 2\Phi\big(t/\sqrt{2}\big) - 1$$

where Φ denotes the standard normal cumulative distribution function. The population median m of D satisfies

$$2\Phi\big(m/\sqrt{2}\big) - 1 = \tfrac{1}{2}$$

Solving for m yields

$$m = \sqrt{2}\,\Phi^{-1}(0.75)$$

The multiset of gaps $\{|X_i - X_j| : i < j\}$ forms a bounded-kernel U-statistic of degree 2. U-statistic consistency results imply that its empirical distribution converges almost surely to the law of D. The sample median of pairwise differences thus converges in probability to m. Convergence in probability plus uniform integrability yields convergence of expectations:

$$\lim_{n\to\infty} \mathbb{E}\big[\text{Spread}(X_1, \ldots, X_n)\big] = \sqrt{2}\,\Phi^{-1}(0.75)$$

Numerically, this equals

$$\lim_{n\to\infty} \mathbb{E}\big[\text{Spread}(X_1, \ldots, X_n)\big] \approx 0.953\,873$$

3.2 Asymptotic Gaussian Efficiency of the Center

This study shows that Center achieves $3/\pi \approx 95.5\%$ efficiency relative to the sample mean under normality. This high efficiency allows Center to handle outliers while maintaining near-optimal performance on normal data.

The analysis uses U-statistic theory to derive the asymptotic distribution of Center under Gaussian data. Let

$$X_1, \ldots, X_n \overset{\text{iid}}{\sim} \mathcal{N}(\mu, \sigma^2), \qquad n \geq 2,\ \sigma > 0$$

Center($\mathbf{x}$) has translation invariance, so setting $\mu = 0$ preserves generality. The Walsh averages are

$$W_{ij} = \frac{X_i + X_j}{2}, \qquad 1 \leq i \leq j \leq n$$

Since $X_i + X_j \sim \mathcal{N}(0, 2\sigma^2)$, it follows that

$$W_{ij} \sim \mathcal{N}\big(0, \sigma^2/2\big), \qquad f_W(0) = \frac{1}{\sigma\sqrt{\pi}}$$

The estimator Center($\mathbf{x}$) equals the sample median of the $\binom{n+1}{2}$ Walsh averages. As a U-quantile of degree two, it satisfies

$$0 = \frac{2}{n(n+1)} \sum_{1 \leq i \leq j \leq n} \Big\{ \mathbf{1}\{W_{ij} \leq \text{Center}(\mathbf{x})\} - \tfrac{1}{2} \Big\}$$

U-quantile theory [Sen63] provides the linear expansion

$$\sqrt{n}\ \text{Center}(\mathbf{x}) = \frac{2}{f_W(0)} \frac{1}{\sqrt{n}} \sum_{i=1}^{n} \psi(X_i) + o_{\Pr}(1)$$

where

$$\psi(x) = \Pr\Big\{ \tfrac{x + X_2}{2} \leq 0 \Big\} - \tfrac{1}{2} = \tfrac{1}{2} - \Phi\Big(\tfrac{x}{\sigma}\Big)$$

and Φ is the standard normal cumulative distribution function. Since $X_1/\sigma \sim \mathcal{N}(0, 1)$,

$$\mathbb{V}\big[\psi(X_1)\big] = \int_{-\infty}^{\infty} \Big(\tfrac{1}{2} - \Phi(u)\Big)^2 \varphi(u)\, du = \frac{1}{12}$$

with φ the standard normal probability density function. Substituting $\mathbb{V}[\psi(X_1)]$ and $f_W(0)$ yields

$$\mathbb{V}\big[\sqrt{n}\ \text{Center}(\mathbf{x})\big] = \frac{4 \cdot \frac{1}{12}}{\big(1/(\sigma\sqrt{\pi})\big)^2} = \frac{\pi\sigma^2}{3}$$

so

$$\sqrt{n}\ \text{Center}(\mathbf{x}) \xrightarrow{d} \mathcal{N}\Big(0, \tfrac{\pi\sigma^2}{3}\Big), \qquad n \to \infty$$

The sample mean has asymptotic variance σ^2/n, hence the *asymptotic Gaussian efficiency* of Center is

$$\text{eff}_{\mathcal{N},\infty}(\text{Center}) = \frac{\sigma^2/n}{\pi\sigma^2/(3n)} = \frac{3}{\pi} \approx 0.954\,930$$

Thus matching the mean's precision under normality requires only $1/0.955 \approx 1.05$ times as many observations, a negligible price for Center's 29% breakdown point and much stronger resistance to outliers.

3.3 Asymptotic Gaussian Efficiency of the Median

This study shows that the sample median achieves only $2/\pi \approx 63.7\%$ efficiency relative to the sample mean under normality.

The analysis applies classical asymptotic theory for sample quantiles to derive the limiting distribution of the median under Gaussian data. Consider

$$X_1, \ldots, X_n \overset{\text{iid}}{\sim} \mathcal{N}(\mu, \sigma^2), \qquad n \geq 2,\ \sigma > 0$$

Since the sample median is translation invariant, set $\mu = 0$ without loss of generality. Write

$$M_n = \text{Median}(X_1, \ldots, X_n)$$

For any continuous distribution with density f positive at its median θ, classical theory ([SSH99], [Ser09]) gives

$$\sqrt{n}\,(M_n - \theta) \xrightarrow{d} \mathcal{N}\Big(0, \frac{1}{4f(\theta)^2}\Big)$$

In the normal case $\theta = 0$ and

$$f(0) = \frac{1}{\sigma\sqrt{2\pi}}$$

so the asymptotic variance becomes

$$\frac{1}{4f(0)^2} = \frac{1}{4}\left(\sigma\sqrt{2\pi}\right)^2 = \frac{\pi\sigma^2}{2}$$

Hence

$$\sqrt{n}\,M_n \xrightarrow{d} \mathcal{N}\Big(0, \frac{\pi\sigma^2}{2}\Big), \qquad n \to \infty$$

The sample mean has asymptotic variance σ^2/n, so the *asymptotic Gaussian efficiency* of the median is

$$\text{eff}_{\mathcal{N},\infty}(\text{Median}) = \frac{\sigma^2/n}{\pi\sigma^2/(2n)} = \frac{2}{\pi} \approx 0.637$$

Thus achieving the same precision as the mean under normality requires $1/0.637 \approx 1.57$ times as many observations when using the median. This large efficiency loss shows why the analysis prefers the Hodges–Lehmann Center estimator, which attains about 95.5% efficiency, whenever data are roughly Gaussian but may include outliers.

3.4 Asymptotic Gaussian Efficiency of the Spread

This study shows that Spread achieves approximately 86% efficiency relative to the sample standard deviation under normality.

The analysis uses U-statistic theory to derive the asymptotic distribution of Spread as a scale estimator under Gaussian data. Consider

$$X_1, \ldots, X_n \overset{\text{iid}}{\sim} \mathcal{N}(\mu, \sigma^2), \qquad n \geq 2,\ \sigma > 0$$

Since $\text{Spread}(\mathbf{x}) = \text{Median}\,|X_i - X_j|$ is translation invariant, set $\mu = 0$ without loss of generality and write

$$R_n = \text{Spread}(\mathbf{x})$$

$$m_0 = \sqrt{2}\,\Phi^{-1}(0.75)\,\sigma \approx 0.954\,\sigma$$

Here m_0 denotes the population median of $|X_i - X_j|$ when $X_i \sim \mathcal{N}(0, \sigma^2)$. Letting $D = |X_1 - X_2|$, its density is

$$f_D(t) = \frac{1}{\sigma\sqrt{\pi}}\,\exp\Big(-\frac{t^2}{4\sigma^2}\Big), \qquad t \geq 0$$

with $f_D(m_0) = \sigma^{-1}\pi^{-1/2}\exp(-\Phi^{-1}(0.75)^2/2)$. Treating R_n as a degree-two U-quantile and applying asymptotic theory [Sen63] yields

$$\sqrt{n}\,\big(R_n - m_0\big) \xrightarrow{d} \mathcal{N}\Big(0, \frac{4\,\mathbb{V}[\psi(X_1)]}{f_D(m_0)^2}\Big)$$

where

$$\begin{aligned}\psi(x) &= \Pr\big\{|x - X_2| \leq m_0\big\} - \tfrac{1}{2} \\ &= \Phi\Big(\tfrac{x+m_0}{\sigma}\Big) - \Phi\Big(\tfrac{x-m_0}{\sigma}\Big) - \tfrac{1}{2}\end{aligned}$$

Numerical evaluation of the integral

$$\mathbb{V}[\psi(X_1)] = \int_{-\infty}^{\infty} \psi(x)^2\,\frac{e^{-x^2/(2\sigma^2)}}{\sigma\sqrt{2\pi}}\,dx$$

yields $\mathbb{V}[\psi(X_1)] \approx 0.0266$. Substituting this value and $f_D(m_0)$ into the variance formula gives

$$\mathbb{V}\big[\sqrt{n}\,R_n\big] \approx 0.527\,\sigma^2$$

Because R_n is *not* a consistent estimator of σ, comparisons with the sample standard deviation $\text{StdDev}(\mathbf{x})$ use the rescaled statistic

$$\widehat{\sigma}_{\text{Spread}} = \frac{R_n}{m_0}$$

which *is* consistent. Dividing the variance above by the constant m_0^2 gives

$$\mathbb{V}\big[\sqrt{n}\,\widehat{\sigma}_{\text{Spread}}\big] \approx 0.579\,\sigma^2$$

The optimal Gaussian scale estimator $\text{StdDev}(\mathbf{x})$ has asymptotic variance $\sigma^2/(2n)$, so the *asymptotic Gaussian efficiency* of the (scaled) Spread is

$$\text{eff}_{\mathcal{N},\infty}(\text{Spread}) = \frac{\sigma^2/(2n)}{0.579\,\sigma^2/n} \approx 0.864$$

One needs roughly $1/0.864 \approx 1.16$ times as many observations to match the precision of the sample standard deviation when the data are exactly normal. In exchange, Spread inherits a 29% breakdown point from its U-quantile construction, so moderately extra data provide a substantial increase in robustness.

3.5 Asymptotic Gaussian Efficiency of the Median Absolute Deviation

This study shows that the median absolute deviation (MAD) achieves only about 37% efficiency relative to the sample standard deviation under normality.

The analysis derives the asymptotic distribution of the MAD under Gaussian data using classical theory for sample medians. Consider

$$X_1, \ldots, X_n \overset{\text{iid}}{\sim} \mathcal{N}(\mu, \sigma^2), \qquad n \geq 2,\ \sigma > 0$$

Since both the sample median and the MAD are translation invariant, set $\mu = 0$ without loss of generality and write

$$\text{MAD}_n = \text{Median}\Big(|X_i - \text{Median}(\mathbf{x})| : i = 1, \ldots, n\Big)$$

For a standard normal distribution, the population MAD equals

$$m_0 = \Phi^{-1}(0.75)\,\sigma = c_0\,\sigma$$

$$c_0 = \Phi^{-1}(0.75) \approx 0.674$$

Dividing the empirical MAD by this constant gives a consistent scale estimator:

$$\widehat{\sigma}_{\text{MAD}} = \frac{\text{MAD}_n}{c_0}$$

To find the large-sample variance, observe that $Y_i = |X_i|$ has density

$$g(y) = \frac{2}{\sigma\sqrt{2\pi}} \exp\Big(-\frac{y^2}{2\sigma^2}\Big), \qquad y \geq 0$$

whose median is m_0. A classical result for sample medians of independent draws with continuous positive density at the median ([SSH99], [Ser09]) states

$$\sqrt{n}\big(\text{Median}(Y_1, \ldots, Y_n) - m_0\big) \xrightarrow{d} \mathcal{N}\Big(0, \tfrac{1}{4g(m_0)^2}\Big)$$

Since

$$g(m_0) = \frac{2}{\sigma\sqrt{2\pi}} \exp\Big(-\frac{c_0^2}{2}\Big)$$

the asymptotic variance of $\sqrt{n}\ \mathrm{MAD}_n$ is

$$\mathbb{V}\big[\sqrt{n}\ \mathrm{MAD}_n\big] = \frac{1}{4g(m_0)^2} = \frac{\pi\sigma^2 \exp(c_0^2)}{8}$$

Scaling by $1/c_0$ gives the variance of the consistent estimator:

$$\mathbb{V}\big[\sqrt{n}\,\widehat{\sigma}_{\mathrm{MAD}}\big] = \frac{\pi\sigma^2 \exp(c_0^2)}{8\,c_0^2}$$

The optimal Gaussian scale estimator is the sample standard deviation. It has asymptotic variance of $\sigma^2/(2n)$. The *asymptotic Gaussian efficiency* of the (scaled) MAD is therefore

$$\mathrm{eff}_{\mathcal{N},\infty}(\mathrm{MAD}) = \frac{\sigma^2/(2n)}{\pi\sigma^2 \exp(c_0^2)/(8c_0^2 n)} = \frac{4c_0^2}{\pi\ \exp(c_0^2)}$$

This gives the result:

$$\mathrm{eff}_{\mathcal{N},\infty}(\mathrm{MAD}) \approx 0.367\,523$$

Achieving the same precision as the sample standard deviation under normality requires $1/0.368 \approx 2.7$ times as many observations when using the MAD.

3.6 Finite-Sample Efficiency of Central Tendency Estimators

This study presents finite-sample efficiency values for Center and shows how it performs better than Median across small and medium sample sizes.

The previous studies established asymptotic efficiency values (the limiting behavior as sample size approaches infinity). For the Gaussian distribution, these asymptotic values are:

- Mean: 100% (the most efficient estimator under normality)
- Center: $3/\pi \approx 95.5\%$
- Median: $2/\pi \approx 63.7\%$

Asymptotic theory provides excellent approximations for large samples but can be inaccurate for small n. Finite-sample efficiency captures the actual precision when working with limited data.

Efficiency and Sample Size Requirements

Efficiency quantifies the variance ratio between estimators. For two estimators T_1 and T_2 applied to the same distribution:

$$\mathrm{eff}(T_1 \text{ relative to } T_2) = \frac{\mathbb{V}[T_2]}{\mathbb{V}[T_1]}$$

This ratio directly translates to sample size requirements. An estimator with 80% efficiency needs $100/80 = 1.25$ times as many observations to achieve the same precision as the reference estimator. The Median with its 63.7% asymptotic efficiency requires roughly 1.57 times more data than the Mean under normality.

Monte Carlo Estimation of Finite-Sample Efficiency

Numerical simulation provides exact efficiency values for any sample size. The procedure follows these steps:

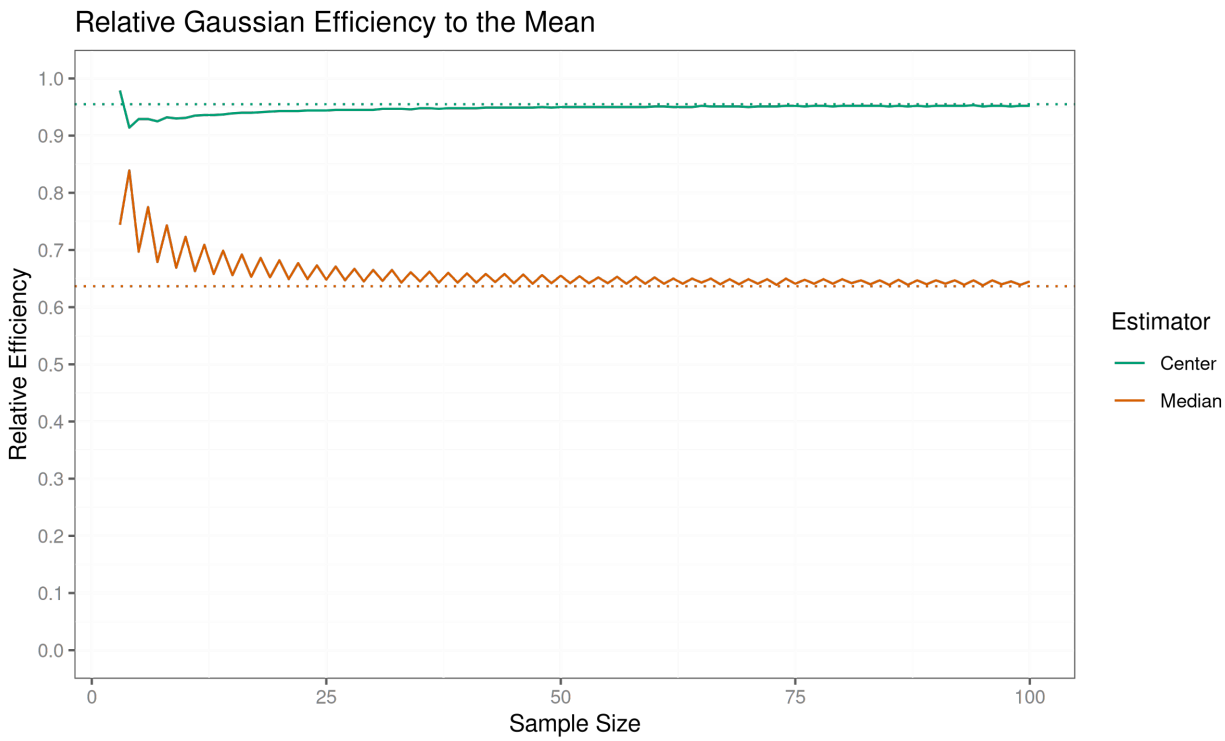
1. **Generate samples**: Draw m independent samples of size n from the standard normal distribution
2. **Calculate estimators**: Compute both estimators for each sample
3. **Measure dispersion**: Calculate the sample variance of the m estimator values
4. **Compute efficiency**: Take the variance ratio

The simulation must balance computational cost against precision. Larger m reduces Monte Carlo error but increases computation time. For efficiency estimation, $m = 10^6$ iterations achieve sufficient precision.

Finite-Sample Results

The simulation reveals how efficiency evolves from small to moderate sample sizes.

The figure below shows the Gaussian efficiency curves for $n \in \{3, \ldots, 100\}$ based on $m = 10^6$ Monte Carlo iterations (dotted lines show asymptotic values):



Key Observations

1. **Small sample behavior**: For $n = 3$, Center achieves 97.9% efficiency while Median drops to 74.3%. Even at extremely small samples, Center maintains above 91% efficiency throughout.

2. **Convergence patterns**: The Center estimator reaches 94% efficiency by $n = 15$ and stabilizes above 95% for $n \geq 50$. The Median oscillates between 63.5% and 68% across all sample sizes, never approaching the efficiency of Center.

3. **Practical advantage**: For typical applications ($n = 20$ to 50), Center maintains $94 - 95\%$ efficiency. This represents only a $5 - 6\%$ penalty compared to Mean, while Median suffers a $35 - 37\%$ penalty.

The toolkit's Center estimator performs much better than Median while sacrificing minimal efficiency compared to the non-robust Mean.

3.7 Finite-Sample Efficiency of Dispersion Estimators

This study presents finite-sample efficiency values for Spread and demonstrates its consistent superiority over the standard deviation StdDev and the median absolute deviation MAD across small and medium sample sizes.

The previous studies established asymptotic efficiency values for dispersion estimators under the Gaussian distribution. These asymptotic values are:

- StdDev: 100% (the most efficient estimator under normality)
- Spread: $\approx 86.4\%$
- MAD: $\approx 36.8\%$

The asymptotic values provide excellent approximations for large samples but may not accurately reflect performance with limited data. Understanding actual efficiency at practical sample sizes guides estimator selection from the toolkit.

Efficiency Measurement for Dispersion Estimators

Dispersion estimators face a variance-bias trade-off: different estimators have different expected values under the same distribution. To compare their variability fairly, the simulation normalizes each estimator by its expected value. This normalization eliminates bias differences and focuses purely on variance comparison.

The procedure follows these steps:

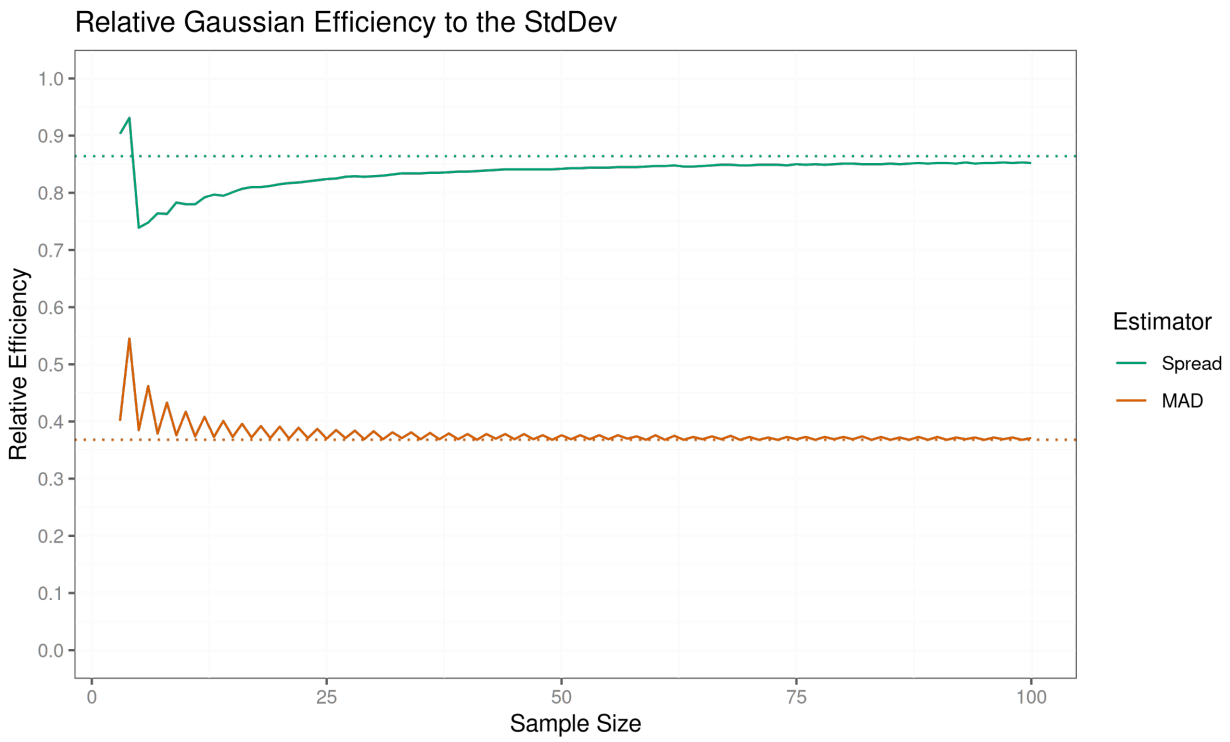
1. **Generate samples**: Draw m independent samples of size n from the standard normal distribution
2. **Calculate estimators**: Compute StdDev, Spread, and MAD for each sample
3. **Calculate normalized variance**: For each set of estimations, calculate the variance and divide by the mean value to align estimator biases
4. **Compute efficiency**: Take the variance ratio relative to StdDev

This approach ensures fair comparison by measuring how much each estimator varies around its own expected value. The simulation uses $m = 10^6$ iterations to achieve sufficient precision.

Results and Analysis

The simulation covers sample sizes $n \in \{3, \ldots, 100\}$ with efficiency computed relative to the standard deviation.

The below figure shows the Gaussian efficiency curves based on 10^6 Monte Carlo iterations (dotted lines show asymptotic values):



Key Observations

1. **Superior efficiency**: Spread consistently outperforms MAD by a factor of two across all sample sizes. At $n = 20$, Spread achieves 81.4% efficiency while MAD reaches only 39.0%.
2. **Practical implications**: For typical sample sizes ($n = 20$ to 50), Spread maintains $81-84\%$ efficiency. This $16-19\%$ penalty compared to StdDev represents a reasonable trade-off for robustness.
3. **Small sample behavior**: Spread shows a characteristic dip for $n = 5-8$ (dropping to 74%) before recovering. Even at its worst, Spread remains twice as efficient as MAD.

Spread from the toolkit provides a substantial efficiency advantage over MAD while maintaining reasonable performance compared to the non-robust standard deviation. This efficiency advantage, combined with its robustness properties, makes Spread the preferred choice for dispersion estimation in practical applications.

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